DISA * LAB

HL7 Specification
Disa*Lab --> CIS

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Laboratory System Technologies

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Contents

1. Introduction

1.1 Overview

An interface is required between the Client Information System(CIS), and the Laboratory System Technologies Disa* Lab Laboratory Information Management System (LIMS). Patient admission/attendance information will be passed from the CIS system to the LIMS, and laboratory results for these patients (where available) will be passed from the LIMS to the CIS system using an electronic interface based on the Health Level 7 (HL7) standard. In addition, the CIS system will also have the ability to order specific tests using this interface. The following principals have been adopted for this interface

a) the interface between the CIS and the LIMS will use the HTTP protocol with message acknowledgements.
b) both the CIS system and the LIMS will populate the HL7 messages to the full extent possible. That is to say, if a data field is available, it will be transmitted by the sending system irrespective of whether or not the receiving system can utilise that data field.
c) For test ordering from the CIS system, the LOINC codes already in use by the laboratories will be used as the coding standard.
d) Two inter-related HL7 specifications will be produced. The CIS system will provide specifications for the HL7 ADT and OML messages, and the LIMS system will provide specifications for the ORU and the ORL messages.
e) any clarification of how a particular HL7 field is to be interpreted will agreed by both the CIS and LIMS vendor, and formally documented in both vendor's specifications.

Please Note: DISA*LAB does not normally send ORL 022 messages. The LLP has sufficient built-in acknowledgement responses.

1.2 Purpose of this document

This document defines the specifications for HL7 messages and segments for general laboratory order response to any OML (ORL eventO22) and for unsolicited transmission of observation records(ORU event R01) from the DISA*LAB Laboratory information management system. This document should be read in conjunction with the specification from CIS for Admissions, Discharges and Transfers (ADT event A01), and for Laboratory Order Messages (OML event O21).

1.3 Lower Layer Protocol

The Lower Layer Protocol (LLP) is the standard for transmitting HL7 messages via HTTP. Each HL7 message must be wrapped using a header and trailer to signify the beginning and end of a message. These headers and footers consist of non-printable characters that are not a part of the actual content of the HL7 message. The overall structure of an HL7 message being sent via LLP is as follows:

Header	HL7 Message	Trailer	Carriage Return
Vertical Tab Character (0x0B)	MSH|^~\&|DISA*LAB|XYZ Laboratory|CIS|ABC HOSPITAL|201904091355||ACK^O21^ACK|2468|P^T|2.8| PID|||MRI123456||LastName^FirstName^MiddleInitial|19800101|M||	Field Separator Character (0x1c)	Carriage Return (0x0D)

1.4 Extraneous Message Segments

Extraneous message segments, such as those that are defined as ignored or not used, should not be included in the HL7 message. When extraneous message segments are present, the segments are ignored and DISA*LAB performs no processing or data validation.

1.5 Separating Components with Delimiters

Certain special characters are used to separate one component from another and are known as delimiter characters. These characters should not be used in the data elements of the HL7 message.

Special Character Name	Special Character
Field Delimiter	|
Component Delimiter	^
Sub-Component Delimiter	&
Repetition Separator	~
Escape Character	\

For each segment of the message, the applicable fields are defined (segment definition tables). Implementation notes are provided to clarify this specific implementation. This is followed by an example.

1.6 References

HL7 Standard, Version 2.5.

1.7 Definitions

(as referenced in the HL7 Standard)

message	A message is the atomic unit of data transferred between systems. It is comprised of a group of segments in a defined sequence.
message type	Defines the purpose of a message. E.g. the ORU message type is used to transmit Unsolicited Observation Report Message data from one system to another
segment	A segment is a logical grouping of data fields. Segments of a message may be required or optional. Each segment is given a name. e.g. ORU message may contain the following segments: Message Header (MSH), Event Type (EVN), Patient ID (PID), Patient Visit (PV1), Observation Report ID (OBR), Observation Common segment (ORC), Observation Result (OBX) and notes (NTE)
battery (profile, panel)	A set of one or more observations identified by a single name and code number, and treated as a shorthand unit for ordering or retrieving results of the constituent observations.
filler (producer)	a person or service who produces the observations (fills the order) requested by the requester, e.g. clinical laboratory system.
placer	a person or service that requests (places order for) an observation battery e.g. HIS order entry or physician's office system

1.8 HL7 hierarchy

- Messages
  - Segments
    - Fields
      - Components
        - Subcomponents

1.9 Conventions

The following defines the conventions used in the segment definition tables (See Section 3).

Segment Attribute	Definition
Seq	(Sequence within the segment) The ordinal position of the data field within the segment. This number is used in the HL7 Standard to refer to the data field in the text comments that follow the segment definition table. Note, the components of a field used by DISA*LAB are listed below the field, for example, as 1.1.
Item#	(ID number) Small integer that uniquely identifies the data item throughout the HL7 Standard.
HL7 description	Descriptive name for the data item used in the HL7 Standard.
Req	(Optionality) Whether the field is required, optional, or conditional in a segment. The designations for optionality are: R = Required O = Optional C = conditional on the trigger event or on some other field(s) X = not used with this trigger event B = left in for backward compatibility with previous version of HL7.
Data type	Restrictions on the contents of the field. Refer to the **HL7 Standard Section 2.5, “Data types”.**
Len	(Length) Maximum number of characters that one occurrence of the data field may occupy.
DISA*LAB implementation	Details of data item as used by the DISA*LAB LIMS.

2. Unsolicited transmission of Observation Messages (ORU/R01)

2.1 Segments

	Segments required/supported	
	MSH	Message header information
	PID	Patient Identification information
	PD1	Patient Demographic - Additional information
	PV1	Patient Visit information
Optional	**NTE**	Notes and comments related to PV1
Conditional if status of order changes	**ORC**	Common Observation
Conditional if results are available	**OBR**	Observation Report ID
Conditional if results are available	**OBX**	Observation Result
Optional	**NTE**	Notes and comments related to OBX

2.2 Example

```
MSH|^~\&|DISA*LAB|University Teaching Hospital|CIS via
kafka||200711201542||ORU^R01^ORU_R01||P^T|2.5|||||ABC
PID|1||ZMP2619003^^^^PI~TYG
88888888^^^^MR~IT2675341^^^^NNABC~A43B21^^^^UU^^^^MA~||BUONAROTI^MICHELANGELO^^^
MR||19750306|M|||FLORENCE^ITALY^^^||076789567|076789300||||||||||||||||N||||||||
|||
PV1|1|I|A4^^^^^^^^A4 Ward|||||^DR BARRETO^|||||||||||1|Patient
Account|||||||||||||||||||||||20070823
NTE|1||COPYDR1:AARCL\F\DR MAX REBO\F\PAAPH\F\Arkham Academy
ORC|SC||||IP|||||||||||||||||||||||||
OBR|1|ZMP2619003||58410-2^Full Blood Count & Platelets^LN^FBCP^Full Blood Count
& Platelets^L|||200708231626|||||^|Gout,
|200708231625|B^^Blood|||||||200708231631|||F||||||Aspirin, Gout medication|||
OBX|1|NM|6690-2^White Cell Count^LN^WCC^White Cell Count^L||10.00|x 10^9/l|
4.00- 10.00||||F|||200708231626|11^ZMP|||
OBX|2|NM|789-8^Red Cell Count^LN^RCC^Red Cell Count^L||5.00|x 10^12/l|  4.50-
5.50||||F|||200708231626|11^ZMP|||
OBX|3|NM|20509-6^Haemoglobin^LN^HB^Haemoglobin^L||14.0|g/dl| 13.0-
17.0||||F|||200708231626|11^ZMP|||
NTE||||Specimen Type||||||||||||||||Specimen Reject Reason||||||||
```

3. DISA*LAB Send Unsolicited Observation Reports (ORU)

An Unsolicited Observation Report Message consists of the following segments:

MSH	Message Header	
PID	Patient Identification	(One patient per message)
PD1	Patient Additional Demographic	(One patient per message)
PV1	Patient Visit Information	(One per message)
OCR	Common Observation	(One per message)
OBR	Observation Request	(Multiple per patient)
OBX	Observation Results	(Multiple per patient)
NTE	Notes & Comments for results	(Multiple per result – refers to the segment which it follows)

3.1 MSH - message header segment

The **MSH** segment defines the intent, source, destination, and some specifics of the syntax of a message,

Example

```
MSH|^~\&|DISA*LAB|ZMP|CIS via
Kafka||200710111505||ORU^R01^ORU_R01|bacb0ebd-cd66-4608-
94ae-acd88df60164|P^T|2.5||||AL|ABC
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00001	2.15.9.1	Field Separator	R	ST	1	'|'
2	00002	2.15.9.2	Encoding characters	R	ST	4	'^~\&'
3	00003	2.15.9.3	Sending Application	O	HD	180, 10	DISA*LAB
3.1			Namespace ID		IS		'DISA*LAB'
4	00004	2.15.9.4	Sending Facility	O	HD	227	Name or three letter code of laboratory performing test.
5	00005	2.15.9.5	Receiving Application	O	HD	227	CIS
6	00006	2.15.9.6	Receiving Facility	O	HD	227	Medical Health facility code or name of facility requesting test.
7	00007	2.15.9.7	Date/Time Of Message	R	TS	26	Date and Time of Message Creation
9	00009	2.15.9.9	Message Type	R	CM	13	Message Type
9.1			Message Code		ID		ORU - Unsolicited observation
9.2			Trigger Event		ID		R01 - Unsolicited observation
9.3			Message Structure		ID		ORU_R01
10	00010	2.15.9.10	Message Control ID	R	ST	199	GUID
11	00011	2.15.9.11	Processing ID	R	PT	3	Processing ID
11.1			Processing ID		ID		P – Production
11.2			Processing Mode		ID		T - Current processing, transmitted at intervals
12	00012	2.15.9.12	Version ID	R	VID	60	HL7 version
12.1			Version ID				2.5
16	00016	2.15.9.16	Application Acknowledgement Type	O	ID	2	See Table 0155 - Accept/application acknowledgment conditions
17	00017	2.15.9.17	Country Code	O	ID	3	3-letter country code (ABC)

Item#	HL7 Standard DISA*LAB	Description
00001	HL7 Standard	**MSH-1 Field separator (ST) 00001** This field contains the separator between the segment ID and the first real field, *MSH-2-encoding characters*. As such it serves as the separator and defines the character to be used as a separator for the rest of the message. Recommended value is |, (ASCII 124).
	DISA*LAB	Uses Recommended value |, (ASCII 124)
00002	HL7 Standard	**MSH-2 Encoding characters (ST) 00002** This field contains the four characters in the following order: the component separator, repetition separator, escape character, and subcomponent separator. Recommended values are ^~\& (ASCII 94, 126, 92, and 38, respectively).
	DISA*LAB	Uses Recommended values ^~\& (ASCII 94, 126, 92, and 38, respectively).
00003	**HL7 Standard**	**MSH-3 Sending application (HD) 00003** This field uniquely identifies the sending application among all other applications within the network enterprise. The network enterprise consists of all those applications that participate in the exchange of HL7 messages within the enterprise.
	DISA*LAB	DISA*LAB
0004	**HL7 Standard**	**MSH-4 Sending facility(HD) 0004** This field further describes the sending application, MSH-3-sending application. With the promotion of this field to an HD data type, the usage has been broadened to include not just the sending facility but other organizational entities such as a) the organizational entity responsible for sending application; b) the responsible unit; c) a product or vendor_s identifier, etc. Entirely site-defined.
	DISA*LAB	Name of Laboratory performing tests e.g. XYZ
0005	**HL7 Standard**	**MSH-5 Receiving Application (HD) 00005** This field uniquely identifies the receiving application among all other applications within the network enterprise. The network enterprise consists of all those applications that participate in the exchange of HL7 messages within the enterprise.
	DISA*LAB	CIS
0006	**HL7 Standard**	**MSH-6 Receiving Facility (HD) 00006** This field identifies the receiving application among multiple identical instances of the application running on behalf of different organization
	DISA*LAB	Name of health facility requesting order. e.g. ABC Hospital
00007	**HL7 Standard**	**MSH-7 Date/time of message (TS) 00007** This field contains the date/time that the sending system created the message. If the time zone is specified, it will be used throughout the message as the default time zone.
	DISA*LAB	Date and time of message creation
00009	**HL7 Standard**	**MSH-9 Message type (CM) 00009** This field contains the message type, trigger event, and the message structure ID for the message. The first component is the message type code defined by HL7 Table 0076 - **Message type.** This table contains values such as ACK, ADT, ORM, ORU etc. The second component is the trigger event code defined by HL7 Table 0003 - **Event type**. This table contains values like A01, O01, R01 etc.
	DISA*LAB	'ORU' - Unsolicited orders, 'R01' - OUL Unsolicited laboratory observation, ORU_R01
00010	**HL7 Standard**	**MSH-10 Message control ID (ST) 00010** This field contains a number or other identifier that **uniquely** identifies the message.

	DISA*LAB	GUID
00011	**HL7 Standard**	**MSH-11 Processing ID (PT) 00011** Components: <processing ID (ID)> ^ <processing mode (ID)> Definition: This field is used to decide whether to process the message as defined in HL7 Application (level 7) Processing rules. The first component defines whether the message is part of a production, training,or debugging system (D=debug, P=production, T=training). The second component defines whether the message is part of an archival process or an initial load (A= Archive, R=restore from archive, I=Initial load, T=Current processing, transmitted at intervals (scheduled or on demand) Not present = the default, meaning current processing). This allows different priorities to be given to different processing modes.
	DISA*LAB	ID value: 'P' – Production Mode value: 'T' - Current processing, transmitted at intervals (scheduled or on demand)
00012	**HL7 Standard**	**MSH-12 Version ID (VID) 00012** This field is matched by the receiving system to its own version to be sure the message will be interpreted correctly.
	DISA*LAB	Version 2.5
00016	**HL7 Standard**	**MSH.16 - Application Acknowledgment Type** This field contains the conditions under which application acknowledgments are required to be returned in response to this message. Required for enhanced acknowledgment mode.
	DISA*LAB	AL. See table 0155 - Accept/application acknowledgment conditions
00017	**HL7 Standard**	**MSH-17 Country Code (ID) 00017** This field contains the country of origin of the message. It will be primarily used to specify default elements, such as currency denominations. The values to be used are those of ISO 3166. HL7 specifies that the 3-character (alphabetic) form be used for the country code.
	DISA*LAB	Value: ABC

3.2 SFT - Software Segment

The **SFT** segment provides additional information about the software product(s) used as a Sending Application. The primary purpose of this segment is for diagnostic use.

Example

```
SFT|Laboratory System Technologies Pty
(Ltd)|05.03.04.1232|DisaHL7Rslt||Disa*Lab|20220726|
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	01834	2.15.12.1	Software Vendor Organization	R	XON	567	Laboratory System Technologies Pty (Ltd)
2	01835	2.15.12.2	Software Certified Version or Release Number	R	ST	15	05.03.04.1232. Version and build will increase as software development occurs.
3	01836	2.15.12.3	Software Product Name	R	ST	20	DisaHL7Rslt
5	01838	2.15.12.5	Software Product Information	O	TX	1024	Disa*Lab
6	01839	2.15.12.6	Software Install Date	O	TS	26	Date of install CCYYMMDD

Item#	HL7 Standard DISA*LAB	Description
00001	HL7 Standard	**SFT.1 - Software Vendor Organization** Organization identification information for the software vendor that created this transaction. The purpose of this field, along with the remaining fields in this segment, is to provide a more complete picture of applications that are sending HL7 messages. The Software Vendor Organization field would allow the identification of the vendor who is responsible for maintaining the application
	DISA*LAB	Laboratory System Technologies Pty (Ltd)
00002	HL7 Standard	**SFT.2 - Software Certified Version or Release Number** Latest software version number of the sending system that has been compliance tested and accepted. Software Certified Version or Release Number helps to provide a complete picture of the application that is sending/receiving HL7 messages.
	DISA*LAB	Current version and build e.g 05.03.04.1232
00003	**HL7 Standard**	**SFT.3 - Software Product Name** The name of the software product that submitted the transaction. A key component in the identification of an application is its Software Product Name. This is a key piece of information in identifying an application.
	DISA*LAB	DisaHL7Rslt
00005	**HL7 Standard**	**SFT.5 - Software Product Information** Software identification information that can be supplied by a software vendor with their transaction. Might include configuration settings, etc.
	DISA*LAB	Disa*Lab
00006	**HL7 Standard**	**SFT.6 - Software Install Date** Date the submitting software was installed at the sending site.
	DISA*LAB	Date when program was compiled

3.3 PID - patient identification segment

The PID segment is used by all applications as the primary means of communicating patient identification information. This segment contains permanent patient identifying and demographic information that, for the most part, is not likely to change frequently, e.g.

Example:

```
PID|1||ZMP2619003^^^^PI~TYG
88888888^^^^MR~IT2675341^^^^NNABC~A43B21^^^^UU^^^^MA~||BUONAR
OTI^MICHELANGELO^^^MR||19750306|M|||FLORENCE^ITALY^^^||076789
567|076789300||||||||||||||||N||||||||||
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00104	3.4.2.1	Set ID - PID	O	SI	4	Only one per patient
3	00106	3.4.2.3	Patient Identifier List	R	CX	250	List of patient identifier numbers.
3.1			ID		ST		Patient identifier number
3.5			Identifier type code		ID		Code identifying the type of number – subset of HL7 Table 0203
5	00108	3.4.2.8	Patient name	R	XPN	250	Patient name
5.1			Family name		FN		Surname
5.2			Given name		ST		First Name
5.5			Prefix		ST		Used if supplied
7	00110	3.4.2.7	Date/Time of Birth	O	TS	26	Date of Birth CCYYMMDD
8	00111	3.4.2.8	Administrative Sex	O	IS	1	Patient's Sex
11	00114	3.4.2.11	Patient Address	O	XAD	250	Patient Address/ Account address
11.1			Street address		ST		Street address/ Postal box
11.2			Other designation		ST		2nd line of address
11.3			City		ST		City
11.4			State or province		ST		Province if given
11.5			Zip or postal code		ST		Post code
13	00116	3.4.2.13	Phone number - home	O	XTN	250	Home Phone number
14	00117	3.4.2.14	Phone number - business	O	XTN	250	Business phone number
30	00741	3.4.2.30	Patient Death Indicator	O	ID	1	Death indicator

Item#	HL7 Standard DISA*LAB	Description
00104	HL7 Standard	**PID-1 Set ID - PID (SI) 00104** This field contains the number that identifies this transaction. For the first occurrence of the segmen sequence number shall be one, for the second occurrence, the sequence number shall be two, etc.
	DISA*LAB	Only one PID is sent per patient.
00106	HL7	**PID-3 Patient identifier list (CX) 00106**

Item#	HL7 Standard DISA*LAB	Description
	Standard	**Components: <ID (ST)> ^ <check digit (ST)> ^ <code identifying the check digit scheme emp (ID)>** ^ < assigning authority (HD)> ^ <identifier type code (ID)> ^ < assigning facility (HD) ^ <effective date (DT)> ^ <expiration date (DT)> Subcomponents of assigning authority: <namespace ID (IS)> & <universal ID (ST)> & <universal ID (ID)> Subcomponents of assigning facility: <namespace ID (IS)> & <universal ID (ST)> & <universal ID ty (ID)> Definition: This field contains the list of identifiers (one or more) used by the healthcare facility to uni identify a patient (e.g., medical record identifier, billing number, birth registry, national unique individ identifier, etc.). Refer also to HL7 Table 0203 - **Identifier type (2.9.12.8)** and User-defined Table 0363 - **Assigning authority** for valid values.
	DISA*LAB	Any or all of the following identifying numbers: Labno (PI - Patient internal identifier) Locn,HospNo (MR – Medical record identifier) ID number (NNABC –National Person identifier - Zambia ID No) Reference Number/s (U – unspecified) Region (RRI - Regional registry ID) Medical Aid (NH – National Health Plan Identifier) Medical Aid no (MA – Medicaid number) UniqueID – Patient matching identifier Original Labno used for comms (PT – Patient external identifier)
00108	**HL7 Standard**	**PID-5 Patient name (XPN) 00108** Components: <family name (FN)> ^ <given name (ST)> ^ <second and further given names or initia thereof (ST)> ^ <suffix (e.g., JR or III)(ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ < type code (ID) > ^ <name representation code (ID)> ^ <name context (CE)> ^ <name validity range ^ <name assembly order (ID)> This field contains the names of the patient, the primary or legal name of the patient is reported first
	DISA*LAB	Patient name (Surname^Name)
00110	**HL7 Standard**	**PID-7 Date/time of birth (TS) 00110** This field contains the patient's date and time of birth.
	DISA*LAB	Only date of birth is used – CCYYMMDD
00111	**HL7 Standard**	**PID-8 Administrative sex (IS) 00111** This field contains the patient's sex.
	DISA*LAB	Defined values: F (Female), M (Male), I (Indeterminate), N (Not applicable)
00114	**HL7 Standard**	**PID-11 Patient address (XAD) 00114** Components: <street address (ST)> ^ <other designation (ST)> ^ <city (ST)> ^ <state or province (ST)> ^ <zip or postal code (ST)> ^<country (ID address type (ID)> ^ <other geographic designation (ST)> ^ <county <address validity range (DR)> Subcomponents of street address: <street address (ST)> & <street name (ST)> & <dwelling numbe (ST)>

Item#	HL7 Standard DISA*LAB	Description
		Definition: This field contains the mailing address of the patient. Address type codes are defined by Table 0190- **Address type**. Multiple addresses for the same person may be sent in the following sequence: The primary mailing address must be sent first in the sequence (for backward compatibil the mailing address is not sent, then a repeat delimiter must be sent in the first sequence.
	DISA*LAB	Patients postal address from accounting information or registration address, if no account number.
00116	**HL7 Standard**	**PID-13 Phone number - home (XTN) 00116** Components: [NNN] [(999)]999-9999 [X99999] [B99999] [C any text] ^ <telecommunication use cod (ID)> ^ <telecommunication equipment type (ID)> ^ <e-mail address (ST)> ^ <country code (NM)> ^ <area/city code (NM)> ^ <phone number (NM)> ^ <extension (NM)> ^ <any text (ST)> This field contains the patient's personal phone numbers. All personal phone numbers for the patier sent in the following sequence. The first sequence is considered the primary number (for backward compatibility). If the primary number is not sent, then a repeat delimiter is sent in the first sequence.
	DISA*LAB	Patients home telephone number
00117	**HL7 Standard**	**PID-14 Phone number - business (XTN) 00117** Components: [NNN] [(999)]999-9999 [X99999] [B99999] [C any text] ^ <telecommunication use cod (ID)> ^ <telecommunication equipment type (ID)> ^ <e-mail address (ST)> ^ <country code (NM)> ^ <area/city code (NM)> ^ <phone number (NM)> ^ <extension (NM)> ^ <any text (ST)> This field contains the patient's business telephone numbers. All business numbers for the patient are sent in the following sequence. The first sequence is considered the patient's primary bu phone number (for backward compatibility). If the primary business phone number is not sent, then a repeat delimiter must be sent in the first sequence.
	DISA*LAB	Patients business telephone number
00741	**HL7 Standard**	**PID-30 Patient death indicator (ID) 00741** This field indicates whether the patient is deceased. Refer to HL7 Table 0136 – *Yes/no indicator* for values.
	DISA*LAB	'Y' – Yes, 'N' – No

3.4 PV1 - patient visit segment

The PV1 segment is used by Registration/Patient Administration applications to communicate information on an account or visit-specific basis.

Example

```
PV1|1|I|A4^^^^^^^^A4 Ward||||^DR BARRETO^||||||||||1|Patient
Account||||||||||||||||||||||||20070823
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00131	3.4.3.1	Set ID = PV1	O	SI	4	1
2	00132	3.4.3.2	Patient Class	R	IS	1	Patient class
3	00133	3.4.3.3	Assigned Patient Location	O	PL	80	Ward
3.1			Point of care		IS		Ward code
3.9			Location description		ST		Ward description
5	00135	3.4.3.5	Pre-admit Number	O	CX	250	Authorization number if supplied
8	00138	3.4.3.8	Referring Doctor	O	XCN	250	Referring Doctor
8.1			ID number		ST		Doctor code (from DISA*LAB dictionaries), could be blank if free text doctor used.
8.2			Family name		ST		Referring doctor's name
8.3			Given name		ST		Doctor HPCSA number (MP number)
19	00149	3.4.3.19	Visit Number	O	CX	250	Visit Number
20	00150	3.4.3.20	Financial Class	O	FC		
44	00174	3.4.3.44	Admit Date/Time	O	TS	26	Admission Date

Item#	HL7 Standard DISA*LAB	Description
00131	HL7 Standard	**PV1-1 Set ID - PV1 (SI) 00131** This field contains the number that identifies this transaction. For the first occurrence of the segment, the sequence number shall be one, for the second occurrence, the sequence number shall be two, etc.
	DISA*LAB	Only 1 per patient. Value: '1'
00132	HL7 Standard	**PV1-2 Patient class (IS) 00132** This field is used by systems to categorize patients by site. It does not have a consistent industry-wide definition. It is subject to site-specific variations. Refer to *User-defined Table 0004 - Patient class* for suggested values.
	DISA*LAB	'I' – Inpatient, 'O' – Outpatient, 'U' – unknown
00133	**HL7 Standard**	**PV1-3 Assigned patient location (PL) 00133** Components: <point of care (IS)> ^ <room (IS)> ^ <bed (IS)> ^ <facility (HD)> ^ <location status (IS)> ^ <person location type (IS)> ^ <building (IS)> ^ <floor (IS)> ^ <location description (ST) Subcomponents of facility: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> This field contains the patient’s initial assigned location or the location to which the patient is being moved.

	DISA*LAB	Ward Code and Description (ABC^^^^^^^^ABC Ward)
00138	**HL7 Standard**	**PV1-8 Referring doctor (XCN) 00138** Components: <ID number (ST)> ^ <family name (ST)> ^ <given name (ST)> ^ <second and further given names or initials thereof (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <source table (IS)> ^ <assigning authority (HD)> ^ <name type code (ID)> ^ <identifier check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ <identifier type code (IS)> ^ <assigning facility (HD)> ^ <name representation code (ID)> ^ <name context (CE)> ^ <name validity range (DR)> Subcomponents of family name: <family name (ST)> & <own family name prefix (ST)> & <own family name (ST)> & <family name prefix from partner/spouse (ST)> & <family name from partner/spouse (ST)> Subcomponents of assigning authority: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> Subcomponents of assigning facility: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> This field contains the referring physician information. Multiple names and identifiers for the same physician may be sent. The field sequences are not used to indicate multiple referring doctors. The legal name must be sent in the first sequence. If the legal name is not sent, then a repeat delimiter must be sent in the first sequence. Depending on local agreements, either the ID or the name may be absent from this field.
	DISA*LAB	Referring doctor code, name, MP number. The doctor's name, MP number will be taken from the doctor's dictionary, if a doctor code was entered, otherwise the doctor's name & MP number will be taken from the registration screen.
00149	**HL7 Standard**	**PV1-19 Visit number (CX) 00149**
	DISA*LAB	Patient visit number
00150	**HL7 Standard**	**PV1.20.1 Financial Class Code**
	DISA*LAB	Patient Account
00174	**HL7 Standard**	**PV1-44 Admit date/time (TS) 00174** This field contains the admit date/time.
	DISA*LAB	Admission Date CCYYMMDD (from v16.04)

3.5 ORC – Common Order Segment

The Common Order segment (ORC) is used to transmit fields that are common to all orders (all types of services that are requested). The ORC segment is required in the Order (ORM) message. ORC is mandatory in Order Acknowledgement (ORR) messages if an order detail segment is present, but is not required otherwise.

Example:

ORC|SC|AABC32149Z|ZMP2619003||IP||||||||||Reason if cancelled||||||||||||

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00215	4.5.1.1	Order Control	R	ID	2	SC (Status change)
2	00216	4.5.1.2	Placer Order number	C	EI	22	Placer Order Number
3	00217	4.5.1.3	Filler Order Number	C	EI	22	DisaLab Laboratory number
5	00219	4.5.1.5	Order Status	O	ID	2	(In process, scheduled)
16	00230	4.5.1.16	Order Control code Reason	O	CE	250	Reason if cancellation

Item#	HL7 Standard DISA*LAB	Description
00215	**HL7 Standard**	**ORC-1 Order Control (ID) 00215** Determines the function of the order segment. Refer to HL7 Table 0119 - Order Control Codes And Their Meaning for valid entries. Depending on the message, the action of the control code may refer to an order or an individual service
	DISA*LAB	Either SC – Status changed or OC -order cancelled
00216	**HL7 Standard**	**ORC.2 - Placer Order Number (ID) 00216** This field is the placer application's order number.
	DISA*LAB	Placer's order no.
00217	**HL7 Standard**	**ORR-3 Filler Order Number (EI) 00217** This field is the order number associated with the filling application.
	DISA*LAB	Laboratory Number
00219	**HL7 Standard**	**ORC-5 Order Status (ID) 00219** This field specifies the status of an order. Refer to HL7 Table 0038 - Order status for valid entries. The purpose of this field is to report the status of an order either upon request (solicited), or when the status changes (unsolicited). It does not initiate action. It is assumed that the order status always reflects the status as it is known to the sending application at the time that the message is send.
	DISA*LAB	If ORC-1 Order control (ID) is SC, then anyone of the following: A – Some, but not all, results available CA – Order was cancelled CM – Order is completed SC – In process, scheduled
00230		**ORC-16 Order Control Code Reason (CE) 00230** This field contains the explanation (either in coded or text form) of the reason for the

	order event described by the order control code (HL7 Table 0119). Whereas an NTE after the order-specific segment (e.g., RXO, ORO, OBR) would provide a comment for that specific segment, the purpose of the order control code reason is only to expand on the reason for the order event. In the case of a cancelled order, for example, this field is commonly used to explain the cancellation
DISA*LAB	Cancellation Reason, If ORC-5 is ‘CA’ the reason will be ‘Rejected’

Please Note: DISA*LAB will only include this segment if the Status of the order is changed in the laboratory.

3.6 OBR – Observation request segment

The Observation Request (OBR) segment is used to transmit information specific to an **order** for a diagnostic study or observation, physical exam, or assessment. Multiple observations can be ordered on a single order segment.

In the **reporting** of clinical data, the OBR serves as the report header. It includes the relevant ordering information when that applies.

```
Example

OBR|1|CIS O/N|ZMP2619003|58410-2^Full Blood Count &
Platelets^LN^FBCP^Full Blood Count &
Platelets^L|||200708231626|||||Gout,
|200708231625|B^Blood||||||200708231631|||F|||||Aspirin, Gout
medication|||
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00237	7.4.1.1	Set ID – OBR	O	SI	4	Set ID - OBR
2	00216	7.4.1.2	Placer Order Number	C	EI	10	Placer Order Number
3	00217	7.4.1.3	Filler Order Number	C	EI	10	Filler Order Number
4	00238	7.4.1.4	Universal Service Identifier	R	CE	250	Test/ Method
4.1			Identifier		ST		LOINC code
4.2			Text		ST		Test/ Method name
4.3			Name of coding System		ID		Identifies the coding scheme being used in the identifier component. Refer to HL7 Table 0396 – Coded values for HL7 tables.
4.4			Alternate Identifier		ST		Local test code
4.5			Alternate text		ST		Test/ Method name
4.6			Name of alternate coding system		ID		Identifies the coding scheme being used in the identifier component. Refer to HL7 Table 0396 – Coded values for HL7 tables.
7	00241	7.4.1.7	Observation Date/Time	C	TS	26	This field is the clinically relevant date/time of the observation.
12	00246	7.4.1.12	Danger Code	O	CE	250	Infection/ precaution
12.2			Text		ST		Description
13	00247	7.4.1.13	Relevant Clinical information	O	ST	300	Clinical information
14	00248	7.4.1.14	Specimen Received Date/Time	C	TS	26	Date and time the specimen was received.
14.1			Date/Time				YYYYMMDDHHMM
15	00249	7.4.1.15	SpecimenSource	O	CM	300	Type of specimen
15.1			Specimen source name or code		CE		Specimen code
15.3			Free text		TX		Specimen code description followed by free text specimens
16	00226	7.4.1.16	Ordering Provider	O	XCN	250	Name of provider who ordered the test

18	00251	7.4.1.18	Placer Field 1	O	ST	60	For future reference
19	00252	7.4.1.19	Placer Field 2	O	ST	60	For future reference
20	00253	7.4.1.20	Filler Field 1	O	ST	60	For future reference
21	00254	7.4.1.21	Filler Field 1	O	ST	60	For future reference
22	00255	7.4.1.22	Results rpt/ status change date/time	O	TS	26	Authorised date and time
25	00258	7.4.1.25	Result Status	C	ID	1	Result status, F=Final Result
26	00259	7.4.1.26	Parent result	O	PRL	400	Used in microbiology for sensitivities
28	00260	7.4.1.28	Result Copies To	O	XCN	250	For future use
31	00263	7.4.1.31	Reason for Study	O	CE	250	Therapy
31.1			Identifier		ST		Therapy patient is on
32	00264	7.4.1.32	Principal Result Interpreter+	O	CM	200	Name of person who authorized the result
32.1			Name		CN		Authorise name
34	00266	7.4.1.34	Technician +	O	CM	200	Name of person who tested/ entered the result
34.1			Name		CN		Tester initials

+ items in this segment are not created by the placer known to the filler, not the placer. They are created by the filler and valued as needed when the OBR segment is returned as part of a report. Hence on a new order sent to the filler, they are not valued.
* fields are only relevant when an observation is associated with a specimen. These are completed by the placer when the placer obtains the specimen. They are completed by the filler when the filler obtains the specimens

Item#	HL7 Standard DISA*LAB	Description
00237	**HL7 Standard**	**OBR-1 Set ID - OBR (SI) 00237** For the first order transmitted, the sequence number shall be 1; for the second order, it shall be 2; and so on.
	DISA*LAB	Number of current order
00216	**HL7 Standard**	**OBR-2 Placer Order Number - OBR (EI) 00216** This field is the placer application's order number.
	DISA*LAB	CIS Order Number
00217	**HL7 Standard**	**OBR-3 Filler Order Number (EI) 00217** This field is the order number associated with the filling application.
	DISA*LAB	Laboratory Number
00238	**HL7 Standard**	**OBR-4 Universal service identifier (CE) 00238** Components: <identifier (ST)> ^ <text (ST)> ^ <name of coding system (IS)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (IS)> This field is the identifier code for the requested observation/test/battery. This can be based on local and/or “universal” codes. We recommend the “universal” procedure identifier.
	DISA*LAB	LOINC code, test method name, test code
00241	**HL7 Standard**	**OBR-7 Observation date/time (TS) 00241** This field is the clinically relevant date/time of the observation. In the case of observationstaken directly from a subject, it is the actual date and time the observation was obtained. In the case of a specimen-associated study, this field shall represent the date and time the specimen was collected or obtained. (This is a results-only field except when the placer or a third party has already drawn the specimen.) This field is conditionally required. When the OBR is transmitted as part of a report message, the field **must** be filled in. If it is transmitted as part of a request

		and a sample has been sent along as part of the request, this field must be filled in because this specimen time is the physiologically relevant date/time of the observation.
	DISA*LAB	Taken date and time, CCYYMMDDHHMM
00246	**HL7 Standard**	**OBR-12 Danger code (CE) 00246** Components: <identifier (ST)> ^ <text (ST)> ^ <name of coding system (IS)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (IS)> This field is the code and/or text indicating any known or suspected patient or specimen hazards, e.g., patient with active tuberculosis or blood from a hepatitis patient. Either code and/or text may be absent. However, the code is always placed in the first component position and any free text in the second component. Thus, free text without a code must be preceded by a component delimiter.
	DISA*LAB	Infectious or precaution alert linked to this specimen
00247	**HL7 Standard**	**OBR-13 Relevant clinical information (ST) 00247** This field contains any additional clinical information about the patient or specimen. This field is used to report the suspected diagnosis and clinical findings on requests for interpreted diagnostic studies.
	DISA*LAB	Clinical information received with specimen
00248	**HL7 Standard**	**OBR-14 Specimen received date/time (TS) 00248** For observations requiring a specimen, the specimen received date/time is the actual login time at the diagnostic service. This field must contain a value when the order is accompanied by a specimen, or when the observation required a specimen **and** the message is a report.
	DISA*LAB	Date and time specimen was received / registered.
00249	**HL7 Standard**	**OBR-15 Specimen source (CM) 00249** Components: <specimen source name or code (CE)> ^ <additives (TX)> ^ <freetext (TX)> ^ <body site (CE)> ^ <site modifier (CE)> ^ <collection method modifier code (CE)> Subcomponents of specimen source name or code: <identifier (ST)> & <test (ST)> & <name of coding system (IS)> & <alternate identifier (ST)> & <alternate text (ST)> & <name of alternate coding system (ST)> This field identifies the site where the specimen should be obtained or where the service should be performed.
	DISA*LAB	Specimen description
00226	**HL7 Standard**	**OBR-16 Ordering Provider (XCN) 00226** Components: <ID Number (ST)> ^ <Family Name (FN)> ^ <Given Name (ST)> ^ <Second and Further Given Names or Initials Thereof (ST)> ^ <Suffix (e.g., JR or III) (ST)> ^ <Prefix (e.g., DR) (ST)> ^ <DEPRECATED-Degree (e.g., MD) (IS)> ^ <Source Table (IS)> ^ <Assigning Authority (HD)> ^ <Name Type Code (ID)> ^ <Identifier Check Digit (ST)> ^ <Check Digit Scheme (ID)> ^ <Identifier Type Code (ID)> ^ <Assigning Facility (HD)> ^ <Name Representation Code (ID)> ^ <Name Context (CE)> ^ <DEPRECATED-Name Validity Range (DR)> ^ <Name Assembly Order (ID)> ^ <Effective Date (TS)> ^ <Expiration Date (TS)> ^ <Professional Suffix (ST)> ^ <Assigning Jurisdiction (CWE)> ^ <Assigning Agency or Department (CWE)> This field identifies the provider who ordered the test. Either the ID code or the name, or both, may be present. This is the same as ORC-12-Ordering provider.
	DISA*LAB	The Referring doctor. I.O.W. The doctor requesting the tests

00251	**HL7 Standard**	**OBR-18 Placer field 1 (ST) 00251** This field is user field #1. Text sent by the placer will be returned with the results.
	DISA*LAB	
00252	**HL7 Standard**	**OBR-19 Placer field 2 (ST) 00252** This field is similar to placer field #1.
	DISA*LAB	For future reference
00253	**HL7 Standard**	**OBR-20 Filler field 1 (ST) 00253** This field is definable for any use by the filler (diagnostic service).
	DISA*LAB	For future reference
00254	**HL7 Standard**	**OBR-21 Filler field 2 (ST) 00254** This field is similar to filler field #1.
	DISA*LAB	For future reference
00255	**HL7 Standard**	**OBR-22 Results Rpt/Status Chng - Date/Time (TS) 00255** This field specifies the date/time results reported or status changed
	DISA*LAB	Authorised date and time
00258	**HL7 Standard**	**OBR-25 Result status (ID) 00258** This field is the status of results for this order. This conditional field is required whenever the OBR is contained in a report message. It is not required as part of an initial order.
	DISA*LAB	F -Final results; results entered and authorised.
00259	**HL7 Standard**	**OBR-26 Parent Result (PRL) 00259** This field is defined to make it available for other types of linkages (e.g., microbiology sensitivities or toxicology).
	DISA*LAB	Microbiology sensitivities
00260	**HL7 Standard**	**OBR-28 Result copies to (XCN) 00260** Components: In Version 2.3 and later, use instead of the CN data type. <ID number (ST)> ^ <family name (FN)> ^ <given name (ST)> ^ <second or further given names or initials thereof (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <source table (IS)> ^ <assigning authority (HD)> ^ <name type code (ID)> ^ <identifier check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ <identifier type code (IS)> ^ <assigning facility (HD)> ^ <name representation code (ID)> ^ <name context (CE)> ^ <name validity range (DR)> ^ < name assembly order (ID)> Subcomponents of assigning authority: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> Subcomponents of assigning facility ID: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> This field is the people who are to receive copies of the results. By local convention, either the ID number or the name may be absent.
	DISA*LAB	Not used – reserved for future use
00263	**HL7 Standard**	**OBR-31 Reason for study (CE) 00263** Components: <identifier (ST)> ^ <text (ST)> ^ <name of coding system (IS)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (IS)> This field is the code or text using the conventions for coded fields given in Chapter 2, Control. This is required for some studies to obtain proper reimbursement.

	DISA*LAB	Description of therapy patient in currently on.
00264	**HL7 Standard**	**OBR-32 Principal result interpreter (CM) 00264** Components: <name (CN)> ^ <start date/time (TS)> ^ <end date/time (TS)> ^ <point of care (IS)> ^ <room (IS)> ^ <bed (IS)> ^ <facility (HD)> ^ <location status (IS)> ^ <patient location type (IS)> ^ <building (IS)> ^ <floor (IS)> Subcomponents of name: <ID number (ST)> & <family name (ST)> & <given name (ST)> & <middle initial or name (ST)> & <suffix (e.g., JR. III) (ST)> & <prefix (e.g., DR)> & <degree (e.g., MD) (ST)> & <source table (IS)> & <assigning authority (HD)> Subcomponents of facility: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> This field identifies the physician or other clinician who interpreted the observation and is responsible for the report content.
	DISA*LAB	Name of person who authorized the results.
00266	**HL7 Standard**	**OBR-34 Technician (CM) 00266** Components: <name (CN)> ^ <start date/time (TS)> ^ <end date/time (TS)> ^ <point of care (IS)> ^ <room (IS)> ^ <bed (IS)> ^ <facility (HD)> ^ <location status (IS)> ^ <patient location type (IS)> ^ <building (IS)> ^ <floor (IS)> Subcomponents of name: <ID number (ST)> & <family name (ST)> & <given name (ST)> & <middle initial or name (ST)> & <suffix (e.g., JR. III) (ST)> & <prefix (e.g., DR)> & <degree (e.g., MD) (ST)> & <source table (IS)> & <assigning authority (HD)> Subcomponents of facility: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> Definition: This field identifies the performing technician.
	DISA*LAB	Name of person responsible for entering test results

3.7 OBX – observation/result segment

The OBX segment is used to transmit a single observation or observation fragment. It represents the smallest indivisible unit of a report.

Example

```
OBX|1|NM|6690-2^White Cell Count^LN^WCC^White Cell
Count^L||10.00|x 10^9/l|  4.00-
10.00||||F|||200708231626|11^ZMP|||
OBX|2|NM|789-8^Red Cell Count^LN^RCC^Red Cell Count^L||5.00|x
10^12/l|  4.50-  5.50||||F|||200708231626|11^ZMP|||
OBX|3|NM|20509-6^Haemoglobin^LN^HB^Haemoglobin^L||14.0|g/dl|
13.0- 17.0||||F|||200708231626|11^ZMP|||
OBX|4|NM|4544-3^HCT^LN^PCV^HCT^L||0.450|l/l| 0.400-
0.500||||F|||200708231626|11^ZMP|||
OBX|5|NM|787-2^MCV^LN^MCV^MCV^L||85.0|fl|  79.1-
98.9||||F|||200708231626|11^ZMP|||
OBX|6|NM|785-6^MCH^LN^MCH^MCH^L||31.0|pg|  27.0-
32.0||||F|||200708231626|11^ZMP|||
OBX|7|NM|786-4^MCHC^LN^MCHC^MCHC^L||35.5|g/dL|  32.0-
36.0||||F|||200708231626|11^ZMP|||
OBX|8|NM|788-0^Red Cell Distribution Width^LN^RDW^Red Cell
Distribution Width^||13.0|%| 11.6-
14.0||||F|||200708231626|11^ZMP|||
OBX|9|NM|777-3^Platelets^LN^PLT^Platelets^L||390|x 10^9/l|
137-  373|H|||F|||200708231626|11^ZMP|||
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00569	7.4.2.1	Set ID – OBX	O	SI	4	Sequence number
2	00570	7.4.2.2	Value Type	C	ID	2	Subset of Value Type
3	00571	7.4.2.3	Observation Identifier	R	CE	250	Test name
3.1			Identifier		ST		LOINC code
3.2			Text		ST		Description or Name of test
3.3			Name of coding system		ID		Identifies the coding scheme being used in the identifier component. Refer to HL7 Table 0396 – Coded values for HL7 tables.
3.4			Alternate Identifier		ST		Local test code
3.5			Alternate Text		ST		Description or Name of test
3.6			Name of alternate coding System		ID		Identifies the coding scheme being used in the identifier component. Refer to HL7 Table 0396 – Coded values for HL7 tables.
4	00572	7.4.2.4	Observation Sub-ID	C	ST	20	Sub-ID
5	00573	7.4.2.8	Observation Value	C	*	65536[1]	Test result
6	00574	7.4.2.6	Units	O	CE	250	Units of measure

7	00575	7.4.2.7	Reference Ranges	O	ST	60	Test reference range
8	00576	7.4.2.8	Abnormal Flags	O	IS	5	Abnormal Flags & Microbiology Sensitivities. Refer to HL7 Table 0078 – **Abnormal flags** for values.
11	00579	7.4.2.11	Observation Result Status	R	ID	1	Result status (F-Final)
14	00582	7.4.2.14	Date/Time of the Observation	O	TS	26	Date/Time of Result Entry
15	00583	7.4.2.15	Producer's ID	O	CE	250	Testing laboratory's region code and prefix
15.1			Identifier		ST		Region number
15.2			Text		ST		Laboratory prefix
18	01479	7.4.2.18	Equipment instance identifier	O	EI	22	Analyser code

[1] The length of the observation field is variable, depending upon value type.

Item#	HL7 Standard DISA*LAB	Description
00569	**HL7 Standard**	**OBX-1 Set ID - OBX (SI) 00569** This field contains the sequence number. For compatibility with ASTM.
	DISA*LAB	Sequential number
00570	**HL7 Standard**	**OBX-2 Value type (ID) 00570** This field contains the format of the observation value in OBX.
	DISA*LAB	**DISA*LAB uses the following:** CE Coded Entry DT Date NM Numeric TM Time TX Text Data ST String Data
00571	**HL7 Standard**	**OBX-3 Observation identifier (CE) 00571** Components: <identifier (ST)> ^ <text (ST)> ^ <name of coding system (IS)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (IS)> This field contains a unique identifier for the observation. The format is that of the Coded Element (CE). Example: 8625-6^P-R interval^LN.
	DISA*LAB	LOINC,Test description, code e.g. 789-9^White Cell Count^LN^WCC^White Cell Count^L
00572	**HL7 Standard**	**OBX-4 Observation sub-ID (ST) 00572** This field is used to distinguish between multiple OBX segments with the same observation ID organized under one OBR.
	DISA*LAB	Used when sending microbiology susceptibility results e.g. OBX|1|CE|SENMI|1|STAAU^Staphylococcus aureus|||||F OBX|2|CE|SENMI|2|ENTEC^Enterococcus species|||||F
00573	**HL7**	**OBX-5 Observation value (*) 00573**

	Standard	This field contains the value observed by the observation producer.
	DISA*LAB	Test (parameter) result
00574	**HL7 Standard**	**OBX-6 Units (CE) 00574** Components: <identifier (ST)> ^ <text (ST)> ^ <name of coding system (IS)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (IS)>
	DISA*LAB	Units as specified in user's dictionaries
00575	**HL7 Standard**	**OBX-7 References range (ST) 00575** Components: for numeric values in the format: d) lower limit-upper limit (when both lower and upper limits are defined, e.g., for potassium 3.5 - 4.5) e) > lower limit (if no upper limit, e.g., >10) f) < upper limit (if no lower limit, e.g., <15)
	DISA*LAB	Reference range as per dictionaries
00576	**HL7 Standard**	**OBX-8 Abnormal flags (IS) 00576** This field contains a table lookup indicating the normalcy status of the result.
	DISA*LAB	Abnormal Flags & Microbiology Sensitivities as per HL7 Table 00780 **– Abnormal flags.** Micro Sensitivities: S=Sensitive, R,=Resistant, I=Intermediate
00579	**HL7 Standard**	**OBX-11 Observation result status (ID) 00579** This field contains the observation result status. Refer to HL7 Table 0085 - **Observation result status codes interpretation** for valid values. This field reflects the current completion status of the results for one Observation Identifier.
	DISA*LAB	Code 'F' – results verified to be correct and final
	HL7 Standard	**OBX-14 Date/time of the observation (TS) 00582** This field is required in two circumstances. The first is when the observations reported beneath one report header (OBR) have different dates/times. It is also needed in the case of OBX segments that are being sent by the placer to the filler, in which case the date of the observation being transmitted is likely to have no relation to the date of the requested observation.
	DISA*LAB	Date and time of analysis/ observation
00583	**HL7 Standard**	**OBX-15 Producer's ID (CE) 00583** Components: <identifier (ST)> ^ <text (ST)> ^ <name of coding system (IS)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (IS)> This field contains a unique identifier of the responsible producing service. It should be reported explicitly when the test results are produced at outside laboratories, for example. When this field is null, the receiving system assumes that the observations were produced by the sending organization.
	DISA*LAB	Testing laboratory region and laboratory prefix
01479	**HL7 Standard**	**OBX-18 Equipment instance identifier (EI) 01479** Components: <entity identifier (ST)> ^ <namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)> This field identifies the Equipment Instance (e.g., Analyzer, Analyzer module, group of Analysers,...) responsible for the production of the observation.
	DISA*LAB	Analyser code if relevant

3.8 NTE – notes and comments

The NTE segment is defined as a common format for sending notes and comments and refers to the segment which it follows.

Example

```
NTE|1||OPTIMAL LIPID PROFILE
NTE|2||Total Cholesterol  <= 5.0 mmol/l
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation DISA*LAB uses the following values:
1	00096	2.16.10.1	Set ID – NTE	O	SI	4	Set ID – NTE
3	00098	2.16.10.3	Comment	O	FT	65536	Comment

Item#	HL7 Standard DISA*LAB	Description
00096	HL7 Standard	**NTE-1 Set ID - NTE (SI) 00096** This field may be used where multiple NTE segments are included in a message. Their numbering must be described in the application message definition.
	DISA*LAB	Sequential number
00098	**HL7 Standard**	**NTE-3 Comment (FT) 00098** This field contains the comment contained in the segment.
	DISA*LAB	Text associated with any segment is added after that segment. **For PV1:** Printed date and time (CCYYMMDDHH:MM) e.g. NTE|3||PRNDT:2007102214:05 Specimen taken and received date and time, and priority (R=Routine, S=Stat, U=Urgent). (TakenDateTime\F\ReceivedDateTime\F\Priority) e.g. NTE|4||SPECDT:2007102210:05\F\RCVDT:2007102211:15\F\Priority:R Registered by, date and time, and modified by, date and time (RegBy\F\RegDateTime\F\ModBy\F\ModDateTime) NTE|5||REGBY:ABC\F\2007102211:15:01\F\MODBY:CBA\F\2007102309:05 **For OBX** Text corresponding to a single line of printed report e.g. NTE|2||Total Cholesterol <= 5.0 mmol/l

4 – General Laboratory order response to any ADT or OML(ORL/O22

After receiving the order request, the pathology laboratory can respond by using the Lab Order Response ORL^O22.The confirmation of receipt as specified in this document contains the following HL7 v2.5 segments:

Segments	Segments required/supported	
	MSH	Message header information
	MSA	Message Acknowledgement
	PID	Patient Identification information
	ORC	Common Order
	ERR	Error Segment
	TQ1	Timing/Quantity
	OBR	Observation Request
	SPM	Specimen

Example

```
MSH|^~\&|DISA*LAB|XYZ Laboratory|CIS|ABC
HOSPITAL|201904091355||ACK^O21^ORL_O22|2468|P^T|2.5||||ABC|||||
MSA|AA|1|||||
PID|||MRN012345||LastName^FirstName^MiddleInitial^Suffix
||19800101|M|||StreetAddress^^City^State^Zip^Country|ABC|HomePhone^^^Email|WorkP
hone|||||||||||||||||||||||||
ORC|OK||||||||||||||||||||||||||||||
ERR||||I|||Your shoe laces are untied||||||
TQ1||||||||||||||
OBR|||||||||||||||||||||||||||||||||
SPM|||||||||||||||||||||||||||||
```

Please Note: DISA*LAB does not normally send ORL 022 messages. The LLP has sufficient built-in acknowledgement responses.

DISA*LAB response to any OML(ORL)

A Laboratory Order Response consists of the following segments:

MSH	Message Header Information	
MSA	Message Acknowledgement	
PID	Patient Identification	(One patient per message)
ORC	Common Order	(One per message)
ERR	Error Segment	(One per message)
TQ1	Timing/Quantity of events	(One per message)
OBR	Observation Request	(One per message)
SPM	Specimen	(One per message)

MSH - message header segment

The Message Header provides sending system and facility information as well as receiving system and agency information. It also contains the message control ID used to uniquely identify each message sent from the facility.

Example

```
MSH|^~\&|DISA*LAB|University Teaching Hospital|CIS via
Kafka||201904091355||ACK^O21^ORL_O22|24G68|P^T|2.5||||ABC||
|||
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00001	2.15.9.1	Field Separator	R	ST	1	'|'
2	00002	2.15.9.2	Encoding characters	R	ST	4	'^~\&'
3	00003	2.15.9.3	Sending Application	O	HD	227	DISA*LAB
4	00004	2.15.9.4	Sending Facility	O	HD	227	Name or three letter code of laboratory performing test.
5	00005	2.15.9.5	Receiving Application	O	HD	227	CIS
6	00006	2.15.9.6	Receiving Facility	O	HD	227	Name of Medical Facility requesting test. Medical Health facility code or .
7	00007	2.15.9.7	Date/Time Of Message	R	TS	26	Date and Time of Message Creation
9	00009	2.15.9.9	Message Type	R	MSG	15	Message Type
9.1			Message Code		ID		ACK -general acknowledgement message
9.2			Trigger Event		ID		O21
9.3			Message Structure		ID		ORL_O22
10	00010	2.15.9.10	Message Control ID	R	ST	20	Random generated 10 alpha-numeric
11	00011	2.15.9.11	Processing ID	R	PT	3	Processing ID
11.1			Processing ID		ID		P - Production
11.2			Processing Mode		ID		T - Current processing, transmitted at intervals
12	00012	2.15.9.12	Version ID	R	VID	60	HL7 version
12.1			Version ID				'2.5'

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
17	00017	2.15.9.17	Country Code	O	ID	3	3 letter Country code

Item#	HL7 Standard DISA*LAB		Description
00001	HL7 Standard		**MSH-1 Field separator (ST) 000**01 This field contains the separator between the segment ID and the first real field, *MSH-2-* *encoding characters*. As such it serves as the separator and defines the character to be used as a separator for the rest of the message. Recommended value is |, (ASCII 124).
	DISA*LAB		Uses Recommended value |, (ASCII 124)
00002	HL7 Standard		**MSH-2 Encoding characters (ST) 00002** This field contains the four characters in the following order: the component separator, repetition separator, escape character, and subcomponent separator. Recommended values are ^~\& (ASCII 94, 126, 92, and 38, respectively).
	DISA*LAB		Uses Recommended values ^~\& (ASCII 94, 126, 92, and 38, respectively).
00003	HL7 Standard		**MSH-3 Sending application (HD) 00003** This field uniquely identifies the sending application among all other applications within the network enterprise. The network enterprise consists of all those applications that participate in the exchange of HL7 messages within the enterprise.
	DISA*LAB		DISA*LAB
0004	HL7 Standard		**MSH-4 Sending facility(HD) 0004** This field further describes the sending application, MSH-3-sending application. With the promotion of this field to an HD data type, the usage has been broadened to include not just the sending facility but other organizational entities such as a) the organizational entity responsible for sending application; b) the responsible unit; c) a product or vendor_s identifier, etc. Entirely site-defined.
	DISA*LAB		Name of Laboratory performing tests e.g. XYZ
0005	HL7 Standard		**MSH-5 Receiving Application (HD) 00005** This field uniquely identifies the receiving application among all other applications within the network enterprise. The network enterprise consists of all those applications that participate in the exchange of HL7 messages within the enterprise.
	DISA*LAB		CIS
0006	HL7 Standard		**MSH-6 Receiving Facility (HD) 00006** This field identifies the receiving application among multiple identical instances of the application running on behalf of different organization
	DISA*LAB		Name of health facility requesting order. e.g. ABC Hospital
00007	HL7 Standard		**MSH-7 Date/time of message (TS) 00007** This field contains the date/time that the sending system created the message. If the time zone is specified, it will be used throughout the message as the default time zone.
	DISA*LAB		Date and time of message creation.
00009	HL7 Standard		**MSH-9 Message type (CM) 00009** This field contains the message type, trigger event, and the message structure ID for the message.

Item#	HL7 Standard DISA*LAB	Description
		The first component is the message type code defined by HL7 Table 0076 - **Message type**. This table contains values such as ACK, ADT, ORM, ORU etc. The second component is the trigger event code defined by HL7 Table 0003 - **Event type**. This table contains values like A01, O01, R01 etc. The third component is the message structure code defined by HL7 Table 0354 – **Message Structure**. This table contains values like A01, O01, R01 etc.
	DISA*LAB	ACK^O21^ORL_O22
00010	**HL7 Standard**	**MSH-10 Message control ID (ST) 00010** This field contains a number or other identifier that uniquely identifies the message. The receiving system echoes this ID back to the sending system in the Message acknowledgement segment (MSA).
	DISA*LAB	Random 10 chars
00011	**HL7 Standard**	**MSH-11 Processing ID (PT) 00011** Components: <processing ID (ID)> ^ <processing mode (ID)> Definition: This field is used to decide whether to process the message as defined in HL7 Application (level 7) Processing rules. The first component defines whether the message is part of a production, training,or debugging system (D=debug, P=production, T=training). The second component defines whether the message is part of an archival process or an initial load (A= Archive, R=restore from archive, I=Initial load, T=Current processing, transmitted at intervals (scheduled or on demand) Not present = the default, meaning current processing). This allows different priorities to be given to different processing modes.
	DISA*LAB	ID value: 'P' – Production Mode value: 'T' - Current processing, transmitted at intervals (scheduled or on demand)
00012	**HL7 Standard**	**MSH-12 Version ID (VID) 00012** This field is matched by the receiving system to its own version to be sure the message will be interpreted correctly.
	DISA*LAB	Version 2.5
00017	**HL7 Standard**	**MSH-17 Country Code (ID) 00017** This field contains the country of origin of the message. It will be primarily used to specify default elements, such as currency denominations. The values to be used are those of ISO 3166. HL7 specifies that the 3-character (alphabetic) form be used for the country code.
	DISA*LAB	Value: ABC

MSA - message acknowledgement

The MSA segment contains information sent while acknowledging another message.

Example

```
MSA|AA|ABC23423DR etc||||
```

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00001	2.15.8.1	Acknowledgement Code	R	ID	2	AA
2	00002	2.15.8.2	Message Control ID	R	ST	20	1
3	00003	2.15.8.3	Text Message	B	ST	80	
4	00004	2.15.8.4	Expected Sequence Number	O	NM	15	
5	00005	2.15.8.5	Delayed acknowledgement Type	W	ID	0	
6	00006	2.15.8.6	Error Condition	B	CE	250	

Item#	HL7 Standard DISA*LAB	Description
1	**HL7 Standard**	**MSA-1 Acknowledgement Code (ID) 00018** This field contains an acknowledgement code, see message processing rules. Refer to HL7 Table 0008- - Acknowledgement code for valid value
	DISA*LAB	AA - Original mode: Application Accept AE - Original mode: Application Error
2	**HL7 Standard**	**MSA-2 Message Control ID (ST) 00010** This field contains the message control ID of the message sent by the sending system. It allows the sending system to associate this response with the message for which it is intended
	DISA*LAB	The GUID from MSH.10 of the received message
3	**HL7 Standard**	**MSA-3 Text Message (ST) 00020** This optional field further describes an error condition. This text may be printed in error logs or presented to an end user The MSA-3 was deprecated as of v 2.4, but is kept in for backward compatibility. The reader is referred to the ERR segment. The ERR segment allows for richer descriptions of the erroneous conditions.
	DISA*LAB	
4	**HL7 Standard**	**MSA-4 Expected Sequence Number (NM) 00021** This optional numeric field is used in the sequence number protocol.

Seq	Item#	HL7 reference	Req	Data type	Len	DISA*LAB implementation

DISA*LAB

5 **HL7 Standard** **MSA-5 Delayed Acknowledgement Type 00022**
The MSA-5 was deprecated as of v2.2 and the detail was withdrawn and removed from the standard as of v 2.5

DISA*LAB

6 **HL7 Standard** **MSA-6 Error Condition (CE) 00023**
This field allows the acknowledging system to use a user-defined error code to further specify AR or AE type acknowledgements.
The MSA-6 was deprecated as of v2.4. The reader is referred to the ERR segment. The ERR segment allows for richer descriptions of the erroneous conditions

DISA*LAB

PID - patient identification segment

The PID segment is used by all applications as the primary means of communicating patient identification information. This segment contains permanent patient identifying and demographic information that, for the most part, is not likely to change frequently, e.g.

Example:

PID|||MRN012345||LastName^FirstName^MiddleInitial^Suffix ||19800101|M|||StreetAddress^^City^State^Zip^ABC||HomePhone^^^Email|WorkPhone||||||||||||||||||||||||

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	00104	3.4.2.1	Set ID - PID	O	SI	4	Only one per patient
3	00106	3.4.2.3	Patient Identifier List	R	CX	250	List of patient identifier numbers.
3.1			ID		ST		CIS patient identifier number
3.5			Identifier type code		ID		Code identifying the type of number – subset of HL7 Table 0203
5	00108	3.4.2.5	Patient name	R	XPN	250	Patient name
5.1			Family name		FN		Surname
5.2			Given name		ST		First Name
5.3			Second and further given names or initials		ST		Not used
5,4			Suffix		ST		Not used
7	00110	3.4.2.7	Date/Time of Birth	R/O	TS	26	Date of Birth CCYYMMDD. Required if know to CIS. If only age is know, then approximate D.O.B. must be calculated, by CIS, and supplied
8	00111	3.4.2.8	Administrative Sex	R/O	IS	1	Patient's Sex. Required if know to CIS
11	00114	3.4.2.114	Patient Address	O	XAD	250	Patient Address/ Account address

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
11.1			Street address		ST		Street address/ Postal box
11.2			Other designation		ST		2nd line of address
11.3			City		ST		City
11.4			State or province		ST		Province if given
11.5			Zip or postal code		ST		Post code
11,6			Country		ID		Country Code
13	116	3.4.2.13	Phone Number - Home	O	XTN	250	Home Phone number
13,4			Email Address		ST		
14	00117	3.4.2.14	Phone number - business	O	XTN	250	Business phone number

Item#	HL7 Standard DISA*LAB	Description
00104	HL7 Standard	**PID-1 Set ID - PID (SI) 00104** This field contains the number that identifies this transaction. For the first occurrence of the segment, the sequence number shall be one, for the second occurrence, the sequence number shall be two, etc.
	DISA*LAB	Only one PID is sent per patient.
00106	HL7 Standard	**PID-3 Patient identifier list (CX) 00106** **Components: <ID (ST)> ^ <check digit (ST)> ^ <code identifying the check digit scheme employed (ID)>** ^ < assigning authority (HD)> ^ <identifier type code (ID)> ^ < assigning facility (HD) ^ <effective date (DT)> ^ <expiration date (DT)> Subcomponents of assigning authority: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)> Subcomponents of assigning facility: <namespace ID (IS)> & <universal ID (ST)> & <universal ID type (ID)>

Definition: This field contains the list of identifiers (one or more) used by the healthcare facility to uniquely identify a patient (e.g., medical record identifier, billing number, birth registry, national unique individual identifier, etc.).
Refer also to HL7 Table 0203 - **Identifier type (2.9.12.5)**
and User-defined Table 0363 - **Assigning authority** for valid values.

DISA*LAB Any or all of the following identifying numbers:

Labno (PI - Patient internal identifier)

Locn,HospNo (MR – Medical record identifier)
ID number (NNABC –National Person identifier - Zambia ID No)
Reference Number/s (U – unspecified)
Region (RRI - Regional registry ID)
Medical Aid (NH – National Health Plan Identifier)
Medical Aid no (MA – Medicaid number)
UniqueID – Patient matching identifier
Original Labno used for comms (PT – Patient external identifier)

00108 **HL7 Standard**

PID-5 Patient name (XPN) 00108
Components: <family name (FN)> ^ <given name (ST)> ^ <second and further given names or initials thereof (ST)> ^ <suffix (e.g., JR or III)(ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <name type code (ID) > ^ <name representation code (ID)> ^ <name context (CE)> ^ <name validity range (DR)> ^ <name assembly order (ID)>

This field contains the names of the patient, the primary or legal name of the patient is
reported first.

DISA*LAB Patient name (Surname^Name)

00110 **HL7 Standard**

PID-7 Date/time of birth (TS) 00110
This field contains the patient’s date and time of birth.

DISA*LAB Only date of birth is used – CCYYMMDD

00111 **HL7 Standard**

PID-8 Administrative sex (IS) 00111
This field contains the patient’s sex.

DISA*LAB Defined values: F (Female), M (Male), I (Indeterminate), N (Not applicable)

00114 **HL7 Standard**

PID-11 Patient address (XAD) 00114
Components: <street address (ST)> ^ <other designation (ST)> ^ <city (ST)> ^ <state or province (ST)> ^ <zip or postal code (ST)> ^<country (ID)> ^ < address type (ID)> ^ <other geographic designation (ST)> ^
<county/parish code (IS)> ^ <census tract (IS)> ^ <address representation code (ID)> ^
<address validity range (DR)>

Subcomponents of street address: <street address (ST)> & <street name (ST)> & <dwelling number (ST)>

Definition: This field contains the mailing address of the patient. Address type codes are defined by HL7 Table 0190 - **Address type**. Multiple addresses for the same person may be sent in the following sequence: The primary mailing address must be sent first in the sequence (for backward compatibility); if the mailing address is not

		sent, then a repeat delimiter must be sent in the first sequence.
	DISA*LAB	Patients postal address from accounting information or registration address, if no account number.
00116	**HL7 Standard**	**PID-13 Phone number - home (XTN) 00116** Components: [NNN] [(999)]999-9999 [X99999] [B99999] [C any text] ^ <telecommunication use code (ID)> ^ <telecommunication equipment type (ID)> ^ <e-mail address (ST)> ^ <country code (NM)> ^ <area/city code (NM)> ^ <phone number (NM)> ^ <extension (NM)> ^ <any text (ST)> This field contains the patient’s personal phone numbers. All personal phone numbers for the patient are sent in the following sequence. The first sequence is considered the primary number (for backward compatibility). If the primary number is not sent, then a repeat delimiter is sent in the first sequence.
	DISA*LAB	Patients home telephone number
00117	**HL7 Standard**	**PID-14 Phone number - business (XTN) 00117** Components: [NNN] [(999)]999-9999 [X99999] [B99999] [C any text] ^ <telecommunication use code (ID)> ^ <telecommunication equipment type (ID)> ^ <e-mail address (ST)> ^ <country code (NM)> ^ <area/city code (NM)> ^ <phone number (NM)> ^ <extension (NM)> ^ <any text (ST)> This field contains the patient’s business telephone numbers. All business numbers for the patient are sent in the following sequence. The first sequence is considered the patient’s primary business phone number (for backward compatibility). If the primary business phone number is not sent, then a repeat delimiter must be sent in the first sequence.
	DISA*LAB	Patients business telephone number

ORC – Common Order Segment

In some cases, the ORC may be as simple as the string ORC|OK|||.

The Common Order segment (ORC) is used to transmit fields that are common to all orders (all types of services that are requested). The ORC segment is required in the Order (ORM) message. ORC is mandatory in Order Acknowledgement (ORR) messages if an order detail segment is present, but is not required otherwise.

Example:

ORC|OK|||||||||||||||||||||||

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
1	215	4.5.1.1	Order Control	R	ID	2	OK (Order Accepted) or UA (Unable to accept order/service)

Item#	HL7 Standard DISA*LAB	Description
215	HL7 Standard	**ORC-1 Order Control (ID) 00215** Determines the function of the order segment. Refer to HL7 Table 0119 - Order Control Codes And Their Meaning for valid entries. Depending on the message, the action of the control code may refer to an order or an individual service
	DISA*LAB	OK or UA. The ERR segment allows for richer descriptions of the erroneous conditions.

Please Note: If ORC.1 contains UA, then the request will **NOT** be processed. Consult the ERR segment to see what the error is to correct it and resubmit.

ERR – Error Segment

The ERR segment is used to add error comments to acknowledgement messages.

Example:

ERR||||W|||your laces are united|||||

Seq	Item#	HL7 reference		Req	Data type	Len	DISA*LAB implementation
4	01815	2.14.5.4	Severity	R	ID	2	W or E
7	01817	2.14.5.7	Diagnostic Information	O	TX	2048	

Item#	HL7 Standard DISA*LAB	Description
01815	HL7 Standard	**ERR-4 Severity (ID) 01814** Identifies the severity of an application error. Knowing if something is Error, Warning or Information is intrinsic to how an application handles the content. Refer to HL7 Table 0516 **-** Error Severity in Chapter 2C, Code Tables, for valid values. If ERR-3 has a value of "0", ERR-4 will have a value of "I".
	DISA*LAB	E(error) Transaction was unsuccessful or W(warning) Transaction successful, but there may issues
01817	HL7 Standard	**ERR-7 Diagnostic Information (TX) 01817** Information that MAY be used by help desk or other support personnel to diagnose a problem.
	DISA*LAB	For an AE code in MSA.1, the error messages, which are in the HL7Ordr log file, will be reflected. e.g 'Errors : Invalid Lab Prefix (MSH.6)' For an AA code in MSA.1, the warning messages, which **might** be in the HL7Ordr log file, will be reflected. e.g ‘Warnings : Specimen Collection DateTime (SPM.18), invalid or not supplied ‘

Please Note: DISA*LAB does not populate TQ1, OBR or SPM segments when sending a ORL message. They are merely provided to maintain the complete structure of the ORL message.